

Problem Statement: Unverified Rural Plantations Suffer 70%+ Mortality & Zero Carbon Accountability

1. WHAT IS THE PROBLEM?

• 70%+ Sapling Mortality:

Over 90% of rural plantation drives lack post-planting tracking; saplings die unnoticed within 12 months.

• Fake Uploads & Greenwashing:

CSR & Govt drives suffer from duplicate stock images and unverified paper claims with zero ground truth.

• Lack of Satellite MRV:

No multi-spectral telemetry exists to audit canopy vigor and tree growth at rural scale.

2. WHO IS AFFECTED & HOW?

• Smallholder & Tribal Farmers:

Deprived of agroforestry yields and excluded from lucrative global voluntary carbon credit markets.

• Funding Agencies (ACIC/CSR/Govt):

100s of crores invested annually with zero proof of tree survival or measurable return on ecology.

• Semi-Arid Ecosystems:

Worsening groundwater depletion, recurring drought cycles, and topsoil loss in Maharashtra belts.

3. OUR PROPOSED SOLUTION

• Sentinel-2 Multi-Spectral Telemetry:

Automated 36-month NDVI & NDRE canopy vigor curves tracking real biomass growth.

• Gemini Edge AI Anti-Fraud Vision:

Instant non-tree auto-rejection + autonomous 70% threshold verification and instant eco-points.

• Digital Passports & Vernacular Voice:

Geo-tagged batch vector QR sheets + Marathi/Hindi audio agro-advisories for rural farmers.

4. POSITIVE CHANGE & BENEFITS

• >90% 3-Year Tree Survival:

AI-driven disease alerts and bio-remedies (NSKE 5%, Trichoderma) prevent sapling die-off.

• Direct Carbon & Livelihood Income:

Enables certified IPCC-compliant carbon credit payouts directly to rural planters.

• 100% Audit-Grade Transparency:

Provides ACIC, NITI Aayog, and CSR sponsors tamper-proof GIS dashboards with cryptographic proof.

50,000 Verified Trees

12-Month Target

>90% Survival Rate

vs 30% Unmonitored Avg

1,000 MT CO2e / yr

Certified Sequestration

¹18L+ ARR Breakeven

Commercial Sustainability